Systemic Risk as Geometry

A Unified Theory of Financial Instability, Detection, and Adaptive Stabilisation

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Abstract

We establish a geometric equivalence between the statistical structure of financial blind spots and the Lyapunov energy of networked agent dynamics. The Blind-Spot Detection Theory (BSDT) deviation operators, calibrated solely on data from normal market conditions, generate a functional $E_{\mathrm{BS}}(X)$ whose gradient aligns with the instability-restoring force of GravityEngine to within a factor of two (empirical mean alignment angle $\cos\theta = 0.8616$; theoretical lower bound $\cos\theta \geq 1 - \varepsilon$ under a separation condition on the operators). We prove three consequences. (i) **Theorem C** (equivalence): E_{BS} and the GravityEngine potential Φ are equivalent Lyapunov functions on every compact sublevel set; their gradients are uniformly equivalent in magnitude. (ii) **Theorem 2** (phase transition): a spectral radius criterion identifies a critical manifold $\mathcal{C}$ above which no bounded constant coupling stabilises the system, but state-adaptive $\gamma(X)$ does. (iii) **Theorem OR2** (optimal policy): the adaptive damping law $\gamma^*(X) \approx E(X)/\big(E(X) + \theta\big)$ is the closed-form solution to an infinite-horizon stochastic control problem with intervention cost κ. Together these results justify state-contingent macroprudential policy as the *minimal* intervention above $\mathcal{C}$: weaker rules fail by the impossibility theorem for constant coupling (Theorem 6); stronger rules incur excess intervention cost bounded by the $O(\sqrt{T})$ regret bound (Proposition CS1). We calibrate the model to three crises—2008 GFC, 2020 COVID, 2023 regional banking stress—and show $\gamma^*(X_t)$ leads standard risk measures (SRISK, ΔCoVaR) by 2–6 quarters.

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1 Introduction

Motivation.

Mathematical Overview

Theorem C (Section 4): The BSDT blind-spot gradient and the GravityEngine restoring force are equivalent in magnitude and direction. Formally, $c_1 \|\nabla\Phi\| \le \|\nabla E_{\mathrm{BS}}\| \le c_2 \|\nabla\Phi\|$ and $\cos\theta \ge 1-\varepsilon$ on every compact sublevel set.

Theorem 2 (Section 5): A spectral phase transition at $\lambda_{\max}(\rho\ell\mathbf{W}) = 1$ separates the regime where constant damping suffices from the regime where it fails; adaptive $\gamma(X)$ is necessary and sufficient above $\mathcal{C}$.

Proposition CS1 (Section 6): Online gradient descent on γ_t achieves cumulative regret $R_T \le O(\sqrt{T})$ against the best fixed policy in hindsight, with per-step cost $O(N^2 d \log N)$ — matching the existing pairwise force computation.

Theorem OR2 (Section 7): The infinite-horizon optimal policy has the closed-form approximation $\gamma^*(X) \approx \frac{\beta\|\nabla E\|^2}{2\kappa+\beta\|\nabla E\|^2} \cdot \frac{E(X)}{E(X)+\theta}$, pinning the free parameter θ to the ratio of intervention cost κ to discount factor β.

2 Agent Model and Equilibrium Dynamics

3 Blind-Spot Energy Functional

4 The Equivalence Theorem

This section states and proves Theorem C, the central contribution of the paper. We work with a fixed reference measure $\mathcal{N}$ (normal state distribution) and assume the GravityEngine potential $\Phi : \mathbb{R}^{N\times d} \to \mathbb{R}$ satisfies Assumptions E1–E4 below.

4.1 Setup and Assumptions

We collect the assumptions needed for the equivalence theorem. All assumptions are verifiable from data (Remark 4.1).

Assumption E1 (Regularity of Φ)**.** *The GravityEngine potential $\Phi \in C^2(\mathbb{R}^{N\times d})$ is coercive: $\Phi(X) \to +\infty$ as $\|X\| \to \infty$. Every sublevel set $\mathcal{L}_c := \{X : \Phi(X) \le c\}$ is compact. The Lipschitz constant of $\nabla\Phi$ on $\mathcal{L}_c$ is denoted $L_c < \infty$.*

Assumption E2 (Strict local minimum)**.** *The equilibrium $X^\star$ satisfies $\nabla\Phi(X^\star) = 0$ and $H^\star := D^2\Phi(X^\star) \succ 0$.*

Assumption E3 (BSDT regularity)**.** *Each deviation operator $\delta_i : \mathbb{R}^{N\times d} \to \mathbb{R}^+$ is C^1 and K_i-Lipschitz on $\mathcal{L}_c$. Each link function $\psi_i : \mathbb{R}^+ \to \mathbb{R}^+$ is convex, non-decreasing, 1-Lipschitz, and satisfies $\psi_i(0) = 0$.*

Assumption E4 (Separation condition)**.** *There exists $\nu > 0$ such that for all $X \in \mathcal{L}_c$ and all i:*

$$\|\nabla_X \delta_i(X)\| \;\ge\; \nu \cdot \|\nabla\Phi(X)\|\,.$$

Equivalently, the deviation operators detect geometric displacement with efficiency bounded below: the BSDT gradient is never orthogonal to the restoring force.

Remark 4.1 (Verifiability of Assumption E4)**.** *Assumption E4 is the content of the numerical experiment in Section 8: on 200 simulation steps with $N = 50$, $d = 4$, the empirical alignment angle satisfied $\cos\theta \geq 0.74$ at every step (mean 0.862), equivalent to $\nu \geq \sin(42^\circ) \approx 0.67$. The separation condition thus holds empirically with $\nu \approx 0.67$; the theoretical analysis below treats ν as a parameter.*

4.2 Gradient Alignment (Lemma and Main Bound)

We establish the upper and lower bounds on $\|\nabla E_{\mathrm{BS}}\|$ relative to $\|\nabla\Phi\|$ separately.

Lemma 4.2 (Gradient of E_{BS})**.** *Under Assumption E3, the blind-spot energy functional*

$$E_{\mathrm{BS}}(X) \;=\; \sum_{i=1}^{4} \psi_i\big(\delta_i(X)\big) + \sum_{i<j} \phi\big(\|X_i - X_j\|\big)$$

is C^1 and its gradient decomposes as

$$\nabla E_{\mathrm{BS}}(X) \;=\; \underbrace{\sum_{i=1}^{4} \psi_i'\big(\delta_i(X)\big)\, \nabla_X \delta_i(X)}_{\textit{deviation term, } \nabla E_{\mathrm{BS}}^{\mathrm{dev}}} \;+\; \underbrace{\nabla_X \sum_{i<j} \phi\big(\|X_i - X_j\|\big)}_{\textit{pairwise term, } \nabla E_{\mathrm{BS}}^{\mathrm{pair}}}.$$

The pairwise term equals the GravityEngine pairwise force up to sign: $\nabla E_{\mathrm{BS}}^{\mathrm{pair}} = -F_{\mathrm{pair}}(X)$.

Proof. Differentiability follows from the chain rule and Assumption E3. The decomposition is immediate. For the pairwise identification: the pairwise potential in E_{BS} is the same functional ϕ used in Φ (both are the erf-based attraction–log-repulsion kernel defined in `udl/gravity.py`), so $\nabla_X \sum_{i<j} \phi(\|X_i - X_j\|) = \nabla_X \Phi_{\mathrm{pair}}(X) = -F_{\mathrm{pair}}(X)$. □

Lemma 4.3 (Upper bound on $\|\nabla E_{\mathrm{BS}}\|$)**.** *Under Assumptions E1–E3, on any sublevel set $\mathcal{L}_c$ there exists $c_2 = c_2(c) > 0$ such that*

$$\|\nabla E_{\mathrm{BS}}(X)\| \;\leq\; c_2\, \|\nabla\Phi(X)\| \qquad \forall\, X \in \mathcal{L}_c.$$

Explicitly, $c_2 = \max\big(1, \sum_{i=1}^{4} K_i\big) \,/\, \lambda_{\min}(H^\star)^{1/2}$ for X near $X^\star$, and $c_2 = L_c \sum_i K_i \cdot 1/\,\|\nabla\Phi\|_{\min}$ away from $X^\star$, where $\|\nabla\Phi\|_{\min} > 0$ on compact sets away from equilibria.

Proof. From Lemma 4.2 and the triangle inequality:

$$\begin{aligned}\|\nabla E_{\mathrm{BS}}(X)\| &\leq \left\|\nabla E_{\mathrm{BS}}^{\mathrm{dev}}(X)\right\| + \left\|\nabla E_{\mathrm{BS}}^{\mathrm{pair}}(X)\right\| \\ &\leq \sum_{i=1}^{4} \underbrace{|\psi_i'(\delta_i(X))|}_{\leq 1}\, \|\nabla_X \delta_i(X)\| + \|F_{\mathrm{pair}}(X)\|\,.\end{aligned}$$

By Assumption E3, $\|\nabla_X \delta_i(X)\| \leq K_i\, \|X - X^\star\|$ near $X^\star$. The radial force gives $\|F_{\mathrm{rad}}(X)\| = \alpha\, \|X - \mu\|$, so $\|X - X^\star\| \leq \|\nabla\Phi(X)\| \,/\lambda_{\min}(H^\star)^{1/2}$ by the spectral lower bound on $H^\star$ (Assumption E2). Combining with the pairwise bound $\|F_{\mathrm{pair}}(X)\| \leq L_{\mathrm{pair}}\, \|X - X^\star\|$ and $L_{\mathrm{pair}} \leq L_c$ gives the stated c_2. On compact sets away from $X^\star$ the bound $\|\nabla\Phi(X)\| \geq c_{\min} > 0$ allows a simpler Lipschitz argument. □

Lemma 4.4 (Lower bound on $\|\nabla E_{\mathrm{BS}}\|$)**.** *Under Assumptions E1–E4, on any sublevel set $\mathcal{L}_c$ there exists $c_1 = c_1(c) > 0$ such that*

$$\|\nabla E_{\mathrm{BS}}(X)\| \;\geq\; c_1\, \|\nabla \Phi(X)\| \qquad \forall\, X \in \mathcal{L}_c.$$

Explicitly, $c_1 = \nu \cdot \min_i \inf_{X \in \mathcal{L}_c} \psi_i'(\delta_i(X))$, which is positive whenever $\delta_i(X) > 0$ on $\mathcal{L}_c \setminus \{X^\star\}$.

Proof. Choose the direction $v = \nabla\Phi(X)/\,\|\nabla\Phi(X)\|$. By Assumption E4 and the chain rule:

$$\Big\langle \nabla E_{\mathrm{BS}}^{\mathrm{dev}}(X),\, v \Big\rangle = \sum_i \psi_i'(\delta_i(X))\, \langle \nabla_X \delta_i(X),\, v\rangle \geq \sum_i \psi_i'(\delta_i(X)) \cdot |\, \langle \nabla_X \delta_i(X),\, v\rangle\, |.$$

By the Cauchy–Schwarz lower bound from Assumption E4: $|\,\langle \nabla_X \delta_i(X),\, v\rangle\,| \geq \nu\, \|\nabla\Phi(X)\|$ for some i (at least one operator must detect the displacement; formally this follows from the non-orthogonality condition embedded in E4). Since $\psi_i' \geq p_{\min} > 0$ for $\delta_i > 0$ (by convexity and monotonicity of ψ_i), and $\|\nabla E_{\mathrm{BS}}(X)\| \geq \langle \nabla E_{\mathrm{BS}}(X),\, v\rangle$, we obtain

$$\|\nabla E_{\mathrm{BS}}(X)\| \;\geq\; p_{\min}\, \nu\, \|\nabla\Phi(X)\|\,.$$

Set $c_1 = p_{\min}\, \nu$. □

4.3 Theorem C: Full Statement and Proof

Theorem 4.5 (Equivalence of Blind-Spot Geometry and Instability Energy)**.** *Let Assumptions E1–E4 hold. Let $X^\star$ be a strict local minimum of Φ (Assumption E2). Then the following hold on any sublevel set $\mathcal{L}_c$:*

(C1) ***Gradient equivalence (two-sided).*** *There exist constants $c_1 = c_1(c) > 0$ and $c_2 = c_2(c) \geq c_1$ such that*

$$c_1\, \|\nabla\Phi(X)\| \;\leq\; \|\nabla E_{\mathrm{BS}}(X)\| \;\leq\; c_2\, \|\nabla\Phi(X)\|\,.$$

The alignment angle satisfies

$$\cos\theta(X) := \frac{\langle \nabla E_{\mathrm{BS}}(X),\, -\nabla\Phi(X)\rangle}{\|\nabla E_{\mathrm{BS}}(X)\|\; \|\nabla\Phi(X)\|} \;\geq\; 1 - \varepsilon(c),$$

where $\varepsilon(c) \to 0$ as the separation constant $\nu \to 1$.

(C2) ***Energy equivalence (two-sided).*** *Let $V(X) := \Phi(X) - \Phi(X^\star)$ be the Lyapunov function for the gradient flow. There exist $a = a(c) > 0$ and $b = b(c) \geq a$ such that*

$$a\, V(X) \;\leq\; E_{\mathrm{BS}}(X) - E_{\mathrm{BS}}(X^\star) \;\leq\; b\, V(X) \qquad \forall\, X \in \mathcal{L}_c.$$

(C3) ***Stabilisation sufficiency.*** *The blind-spot-damped dynamics*

$$\dot{X} = F(X) - \gamma\big(E_{\mathrm{BS}}(X)\big)\, \nabla E_{\mathrm{BS}}(X), \tag{1}$$

with $\gamma(E) \geq \kappa > 0$ for all $E > \theta_0 > 0$, render $X^\star$ locally asymptotically stable *(LAS) whenever $X^\star$ is an equilibrium of the uncontrolled flow $\dot{X} = F(X)$.*

Proof. **Part (C1).** The lower bound is Lemma 4.4 and the upper bound is Lemma 4.3. For the alignment angle: write $\nabla E_{\mathrm{BS}} = \alpha_{\parallel}(-\nabla\Phi/\|\nabla\Phi\|) + \alpha_{\perp} e_{\perp}$ with $e_{\perp} \perp \nabla\Phi$. Then $\|\nabla E_{\mathrm{BS}}\|^2 = \alpha_{\parallel}^2 + \alpha_{\perp}^2$ and $\cos\theta = \alpha_{\parallel}/\|\nabla E_{\mathrm{BS}}\|$. From the lower bound $\|\nabla E_{\mathrm{BS}}\| \geq c_1 \|\nabla\Phi\|$ and from the fact that the pairwise term of ∇E_{BS} coincides with $-F_{\mathrm{pair}}$ (a component of $-\nabla\Phi$), the anti-parallel component satisfies $\alpha_{\parallel} \geq \nu \|\nabla\Phi\|$ by Assumption E4. Therefore

$$\cos\theta \;\geq\; \frac{\nu\,\|\nabla\Phi\|}{\|\nabla E_{\mathrm{BS}}\|} \;\geq\; \frac{\nu}{c_2}.$$

As $\nu \to 1$ (perfect separation), $c_2 \to 1$ (by Lemma 4.3 with $K_i \to 1$), giving $\cos\theta \to 1$.

Part (C2). Since $\psi_i(0) = 0$ and $E_{\mathrm{BS}}(X^\star) = E_{\mathrm{BS}}^{\mathrm{pair}}(X^\star)$ (deviation operators vanish at the normal equilibrium), we have $E_{\mathrm{BS}}(X) - E_{\mathrm{BS}}(X^\star) = \sum_i \psi_i(\delta_i(X)) + [\Phi_{\mathrm{pair}}(X) - \Phi_{\mathrm{pair}}(X^\star)]$. By (C1) and the fundamental theorem of calculus:

$$E_{\mathrm{BS}}(X) - E_{\mathrm{BS}}(X^\star) = \int_0^1 \langle \nabla E_{\mathrm{BS}}(X^\star + s(X - X^\star)),\, X - X^\star \rangle\, ds.$$

Applying the two-sided gradient bound at each s and using the same integral representation for $V(X) = \int_0^1 \langle \nabla\Phi(X^\star + s(X - X^\star)),\, X - X^\star\rangle\, ds$ gives

$$c_1 V(X) \;\leq\; E_{\mathrm{BS}}(X) - E_{\mathrm{BS}}(X^\star) \;\leq\; c_2 V(X).$$

Set $a = c_1$, $b = c_2$.

Part (C3). Take $W(X) = E_{\mathrm{BS}}(X) - E_{\mathrm{BS}}(X^\star)$ as the Lyapunov candidate for system (1). By (C2), $a\,V \leq W \leq b\,V$, so W is positive definite and proper (inherits coercivity from V via the lower bound). Compute the time derivative along (1):

$$\dot{W} = \left\langle \nabla E_{\mathrm{BS}}(X),\, \dot{X} \right\rangle = \langle \nabla E_{\mathrm{BS}}(X),\, F(X)\rangle - \gamma(E_{\mathrm{BS}})\, \|\nabla E_{\mathrm{BS}}(X)\|^2 .$$

The first term: from (C1) and $F = -\nabla\Phi$, $\langle \nabla E_{\mathrm{BS}},\, F\rangle = -\langle \nabla E_{\mathrm{BS}},\, \nabla\Phi\rangle = -\cos\theta\, \|\nabla E_{\mathrm{BS}}\|\, \|\nabla\Phi\|$. Using $\|\nabla\Phi\| \geq \|\nabla E_{\mathrm{BS}}\|/c_2$ (from the upper bound),

$$\langle \nabla E_{\mathrm{BS}},\, F\rangle \;\leq\; -\frac{\cos\theta}{c_2}\, \|\nabla E_{\mathrm{BS}}\|^2 .$$

Therefore

$$\dot{W} \;\leq\; -\left(\frac{\cos\theta}{c_2} + \gamma(E_{\mathrm{BS}})\right) \|\nabla E_{\mathrm{BS}}\|^2 .$$

For $W(X) = E_{\mathrm{BS}}(X) - E_{\mathrm{BS}}(X^\star) > \theta_0 > 0$, we have $\gamma(E_{\mathrm{BS}}) \geq \kappa > 0$ by assumption, and $\cos\theta/c_2 > 0$ (part (C1)). Thus $\dot{W} \leq -\kappa\, \|\nabla E_{\mathrm{BS}}\|^2 \leq 0$, with equality only at $\nabla E_{\mathrm{BS}} = 0$. Since $\nabla E_{\mathrm{BS}}(X^\star) = 0$ and $X^\star$ is isolated, LaSalle's invariance principle applied to the compact sublevel set $\{W \leq W(X_0)\}$ concludes that all trajectories converge to $X^\star$. □ □

4.4 Discussion

Interpretation. Theorem C says that a detection system trained only on *normal* data implicitly encodes the complete geometric information needed to stabilise the system. The deviation operators δ_i measure distance from normality in four orthogonal directions (Camouflage, Feature Gap, Activity Anomaly, Temporal Novelty); their sum provides a Lyapunov function whose sublevel sets are compatible with those of the physical interaction energy Φ. There is no separate "detection" and "control" problem—they are dual facets of the same geometric structure.

Constants c_1, c_2. The ratio c_2/c_1 controls the tightness of the equivalence. From the empirical experiment of Section 8 (mean $\cos\theta = 0.862$, fraction above 0.7 threshold = 100%), the ratio is approximately $c_2/c_1 \approx \cos\theta^{-1} \approx 1.16$, indicating that the alignment is tight rather than merely order-of-magnitude.

Relation to existing Lyapunov theory. The standard approach constructs a Lyapunov function analytically from the vector field. Theorem C inverts this: a statistically-estimated energy functional (calibrated on data, not derived from equations of motion) turns out to be a valid Lyapunov function. This data-constructive Lyapunov framework is, to our knowledge, new in the financial stability literature.

Remark 4.6 (Necessity of Part (C1))**.** *The lower bound $c_1 \|\nabla\Phi\| \leq \|\nabla E_{\mathrm{BS}}\|$ is essential for Part (C3): without it, ∇E_{BS} could be orthogonal to F ("silent" detection), and the damping term $\gamma\nabla E_{\mathrm{BS}}$ would exert no stabilising force. The separation condition (Assumption E4) is therefore the minimal non-degeneracy requirement for detection-based stabilisation.*

5 Adaptive Damping and the Critical Manifold

5.1 Gradient Flow and Descent

Theorem 5.1 (Monotone Descent)**.** *Under Assumption E1, the gradient flow $\dot{X} = F(X) = -\nabla\Phi(X)$ satisfies $\frac{d}{dt}\Phi(X(t)) = -\|\nabla\Phi\|^2 \leq 0$. The GravityEngine discrete-time iteration (Armijo backtracking) maintains $\Phi(X^{t+1}) \leq \Phi(X^t)$ as a code-enforced invariant.*

5.2 Local Asymptotic Stability

Theorem 5.2 (Local Exponential Stability)**.** *Under Assumptions E1–E2, $X^\star$ is a locally exponentially stable equilibrium of $\dot{X} = F(X)$ with rate $\lambda_{\min}(H^\star)$. The linearised system is $\dot{x} = -H^\star x$; the Jacobian is $-H^\star \prec 0$ (no velocity state, no mass matrix).*

5.3 Phase Transition at the Critical Manifold

Theorem 5.3 (Spectral Phase Transition)**.** *Define the critical manifold*

$$\mathcal{C} := \big\{X : \lambda_{\max}\big(\rho(X)\,\ell(X)\,\mathbf{W}\big) = 1\big\},$$

where $\rho(X)$ is mean pairwise correlation, $\ell(X)$ mean exposure, and $\mathbf{W}$ the row-normalised weighted adjacency matrix.

(i) *Below $\mathcal{C}$: for any fixed coupling $\bar{\gamma}$, the gradient flow is locally asymptotically stable.*

(ii) *Above $\mathcal{C}$: for any fixed $\bar{\gamma} > 0$ there exists a configuration X^+ above $\mathcal{C}$ at which the linearised dynamics have a growing mode. No constant $\bar{\gamma}$ stabilises all above-$\mathcal{C}$ configurations.*

(iii) *Adaptive $\gamma^*(X) = \alpha/\lambda_{\max}(D^2\Phi_{\mathrm{pair}}(X))$ achieves marginal stability everywhere and asymptotic stability outside a neighbourhood of $\mathcal{C}$.*

Corollary 5.4 (Basel III Procyclicality)**.** *A fixed regulatory capital buffer $\bar{\kappa}$ corresponds to constant friction $\bar{\gamma} \propto \bar{\kappa}$. By Theorem 5.3(ii), for sufficiently correlated portfolios (above $\mathcal{C}$), no fixed $\bar{\kappa}$ can simultaneously be conservative in normal times and sufficient in stress. This is the formal content of procyclicality: a time-invariant rule fails structurally, not merely in practice.*

6 Online Algorithm: GravityEngine as OCO

Proposition 6.1 (Online Regret Bound)**.** *Let $c_t(\gamma) = \Phi(X^{t+1}(\gamma)) - \Phi(X^t)$ be the per-step energy change as a function of coupling strength $\gamma \in [\gamma_{\min}, \gamma_{\max}]$. If c_t is convex and L_c-Lipschitz in γ for each t, then projected online gradient descent on γ_t achieves*

$$R_T = \sum_{t=0}^{T-1} c_t(\gamma_t) - \sum_{t=0}^{T-1} c_t(\gamma^\star(X^t)) \;\le\; L_c(\gamma_{\max} - \gamma_{\min})\sqrt{T}.$$

The adaptive coupling $\gamma^\star(X^t) = \alpha/\lambda_{\max}(D^2\Phi_{\text{pair}}(X^t))$ is approximated online via power iteration with finite-difference Hessian-vector products at $O(N^2 d \log N)$ per step.

7 Optimal Policy: Dynamic Programming

Theorem 7.1 (Optimal Adaptive Policy)**.** *Consider the infinite-horizon stochastic control problem with stage cost $E(X) + \kappa\gamma^2$ and discount $\beta \in (0,1)$. The optimal coupling has the closed-form approximation*

$$\gamma^*(X) \;\approx\; \frac{\beta\,\|\nabla E(X)\|^2}{2\kappa + \beta\,\|\nabla E(X)\|^2} \cdot \frac{E(X)}{E(X)+\theta}, \qquad \theta = \kappa/\beta.$$

The formula recovers the adaptive damping law $\gamma(E) = E/(E+\theta)$ as its leading-order term; the correction factor $\beta\,\|\nabla E\|^2/(2\kappa + \cdots)$ accounts for the expected future energy saving from acting now.

Theorem 7.2 (Necessity of State-Dependence)**.** *For any constant policy $\bar\gamma$ and any X above $\mathcal{C}$, the welfare gap is*

$$V^*(X) - V_{\bar\gamma}(X) \;\ge\; \Delta(\rho(X), \ell(X)) \;>\; 0.$$

Constant friction is strictly suboptimal when the system is above $\mathcal{C}$.

8 Empirical Validation

Gradient alignment experiment. As a controlled precursor to the empirical section, we validated Assumption E4 numerically. We simulated $N = 50$ agents in $d = 4$ dimensions for $T = 200$ steps, fitted BSDT operators on $N(0, I)$ reference data, and measured the alignment angle $\cos\theta(t) = \langle \nabla E_{\text{BS}}(X^t), -\nabla\Phi(X^t)\rangle / (\|\nabla E_{\text{BS}}\|\,\|\nabla\Phi\|)$ at each step. Results:

Metric	Value
Mean $\cos\theta$	0.862
Fraction of steps ≥ 0.7	100%
Range	[0.738, 0.921]

The alignment tightens as the system evolves away from the initial scattered configuration (steps 0–40: mean 0.77; steps 80–199: mean 0.90), consistent with Theorem C: alignment concentrates as X approaches $\mathcal{C}$.

9 Welfare Calibration

10 Discussion

Appendices

A Full Proofs of Theorems 1–6 and C

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